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Week 5: Structure from Motion

- ✔

Ungraded Plugin: 5.1

Overview of Structure from Motion

4 min
- ✔

Practice Assignment: 5.1

Overview of Structure From Motion Self-Check Quiz

Submitted
- ✔

Ungraded Plugin: 5.2

Structure from Motion Problem

4 min
- ✔

Practice Assignment: 5.2

Structure from Motion Problem Self-Check Quiz

Submitted
- ✔

Reading: 5.3 Video

Correction

10 min
- ✔

Ungraded Plugin: 5.3

Observation Matrix

15 min
- ✔

Practice Assignment: 5.3

Observation Matrix Self-Check Quiz

Submitted

✔

Reading: 5.4 Video

Correction

10 min

<>

Ungraded Plugin: 5.4

Rank of Observation Matrix

10 min

📋

Practice Assignment: 5.4

Rank of Observation Matrix Self-Check Quiz

3h

<>

Ungraded Plugin: 5.5

Tomasi-Kanade Factorization

21 min

📋

Practice Assignment: 5.5

Tomasi-Kanade Factorization Self-Check Quiz

3h

💬

Discussion Prompt: Week 5

Structure from Motion

10 min

💬

Discussion Prompt: Week 5

Questions and Feedback

Week 5 Quiz

Post-Course Survey

5.4 Video Correction

In [Video 5.4 Rank of Observation Matrix](#), from 5:28 to 7:01, the second "=" should be placed with "≤" (see below).

*The revisions are shown in **red** and correspond to either corrections to errors or additions made for clarity.

Important Properties of Matrix Rank

- $Rank(A^T) = Rank(A)$
- $Rank(A_{m \times n} B_{n \times p}) \leq \min(Rank(A_{m \times n}), Rank(B_{n \times p})) \leq \min(m, n, p)$
- $Rank(AA^T) = Rank(A^T A) = Rank(A^T) = Rank(A)$
- $A_{m \times m}$ is invertible iff $Rank(A_{m \times m}) = m$ (Full Rank)

MATH PRIMER

From 7:02 to 8:12, the point 1 and point 2 in the penultimate row of the matrix form in slide should be $\tilde{v}_{2,1}$ and $\tilde{v}_{2,2}$ instead of $\tilde{u}_{2,1}$ and $\tilde{u}_{2,2}$.

*The revisions are shown in **red** and correspond to either corrections to errors or additions made for clarity.

Point 1

Point 2

Point N

Image 1

Image 2

Image F

Image 1

Image 2

Image F

$\tilde{u}_{1,1}$

$\tilde{u}_{2,1}$

$\vdots$

$\tilde{u}_{F,1}$

$\tilde{v}_{1,1}$

$\tilde{u}_{2,1}$

$\vdots$

$\tilde{v}_{F,1}$

$\tilde{u}_{1,2}$

$\tilde{u}_{2,2}$

$\vdots$

$\tilde{u}_{F,2}$

$\tilde{v}_{1,2}$

$\tilde{u}_{2,2}$

$\vdots$

$\tilde{v}_{F,2}$

$\dots$

$\dots$

$\dots$

$\dots$

$\dots$

$\dots$

$\dots$

$\dots$

$\tilde{u}_{1,N}$

$\tilde{u}_{2,N}$

$\vdots$

$\tilde{u}_{F,N}$

$\tilde{v}_{1,N}$

$\tilde{u}_{2,N}$

$\vdots$

$\tilde{v}_{F,N}$

$W_{2F \times N}$

Centroid-Subtracted Feature Points (Known)

$=$

$\begin{bmatrix} i_1^T \\ i_2^T \\ \vdots \\ i_F^T \end{bmatrix}$

$M_{2F \times 3}$

Camera Motion (Unknown)

$[P_1 \ P_2 \ \dots \ P_N]$

$S_{3 \times N}$

Scene Structure (Unknown)

From 8:13 to 9:44, it should be $Rank(W) \leq Rank(MS) \leq \min(3, N, 2F)$ (see below).

*The revisions are shown in **red** and correspond to either corrections to errors or additions made for clarity.

Rank of Observation Matrix

$$\begin{matrix} W & = & M & \times & S \\ 2F \times N & & 2F \times 3 & & 3 \times N \end{matrix}$$

We know:

$Rank(MS) \leq Rank(M)$

$Rank(MS) \leq Rank(S)$

$\Rightarrow Rank(MS) \leq \min(3, 2F)$

$Rank(MS) \leq \min(3, N)$

$\Rightarrow Rank(W) \leq Rank(MS) \leq \min(3, N, 2F)$

Rank Theorem: $Rank(W) \leq 3$

[Tomasi 1992]

Go to next item

✔ Completed

